

Tutorial 8

CSE 112 Computer Organization

Note to TAs: Q1 and Q2 are based on Iron law.

(Time/Program) = (Instruction/Program)*(Cycles/Instruction)*(Time/Cycles)

[instruction / time] = [instruction / clock cycle] * [clock cycle / time]

Q1:

- a) No, Processor B might have much higher IPC (instructions per cycle).
For example, a processor with a lower frequency might be able to execute multiple instructions per cycle while a processor with a higher frequency might only execute one instruction per cycle.
- b) Yes, as the number of instructions is the same.

Q2:

- a) ISA A: 10 instructions/cycle * 500,000,000 cycle/second = 5000 MIPS
- b) ISA B: 2 instructions/cycle * 600,000,000 cycle/second = 1200 MIPS
- c) Don't know. The best-compiled code for each processor may have a different number of instructions.
- d) Run time = Number of instructions * Time taken to execute one instruction
For ISA A, Time taken to execute 1 instruction = $1/5000 \times 10^{-6}$ second
For ISA B, Time taken to execute 1 instruction = $1/1200 \times 10^{-6}$ second
Therefore:
 $1/5000 \times 10^{-6} \times 1000 = 1/1200 \times 10^{-6} \times \text{Instructions for ISA B}$
 $\Rightarrow \text{Instructions for ISA B} = 240 \Rightarrow$ For the time in which 1000 instructions get executed for ISA A, only 240 instructions will be executed for ISA B, meaning ISA B i.e., Processor B is slower.

Reference for Q1 and Q2:

https://safari.ethz.ch/digitaltechnik/spring2020/lib/exe/fetch.php?media=optional_hw3.pdf

Q3:

a.

Instruction	1	2	3	4	5	6	7	8
add r1 r2 r3	F	D	X	M	W			
add r4 r5 r6		F	D	X	M	W		
add r7 r2 r3			F	D	X	M	W	
add r8 r2 r5				F	D	X	M	W

Number of cycles = 8

b. G

Instruction	1	2	3	4	5	6	7	8
mov r1 r3	F	D	X	M	W			
add r3, r2, r4		F	D	X	M	W		
add r5, r2, r4			F	D	X	M	W	
mov r7, r8				F	D	X	M	W

Number of cycles = 8

Q4.

Note to TAs : Do not consider hazards for pipelining. Assume pipeline does not stall for hazards.

With no Pipeline

add r1, r2, r3 // $r1 = 2+3 = 5$
 add r3, r4, r1 // $r3 = 4+5 = 9$
 add r5, r7, r3 // $r5 = 7 + 9 = 16$

With 5 stage Pipeline

add r1, r2, r3 // $r1 = 2+3 = 5$
 add r3, r4, r1 // $r3 = 4+1 = 5$
 add r5, r7, r3 // $r5 = 7 + 3 = 10$

The execution done with 5 stage pipeline continues with the execution and does not consider the updated value of registers. This happens as we have not considered hazard with the pipelining.

Note to TAs: This question is open ended. Students should consider the various possibilities of why is there is mismatch in the values.